Opgaver uge 8

Klassisk Fysik, Wolfson’s kapitel 10+11

Kapitel 10

Problems

45. A wheel turns through 3.0 revolutions while accelerating from rest at 17 rpm/s. (a) What's its final angular speed? (b) How long does it take?

60. A ship's anchor weighs 5.0 kN. Its cable passes over a roller of negligible mass and is wound around a hollow cylindrical drum of mass 380 kg and radius 1.1 m, mounted on a frictionless axle. The anchor is released and drops 16 m to the water. Use energy considerations to determine the drum's rotation rate when the anchor hits the water. Neglect the cable's mass.

66. A 20-kg mass is tied to a massless rope wrapped around a solid cylindrical drum mounted on a frictionless horizontal axle. When the mass is released, it falls with acceleration $a = 5.0\ \mathrm{m/s^2}$. Find (a) the rope tension and (b) the drum's mass.

Kapitel 11

Problems

35. You slip a wrench over a bolt. Taking the origin at the bolt, the other end of the wrench is at $x = 25$ cm, $y = 6.5$ cm. You apply a force $\vec{F} = 62\hat{\imath} - 16\hat{\jmath}$ N to the end of the wrench. What's the torque on the bolt?

54. An anemometer for measuring wind speeds consists of four small cups, each with mass 120 g, mounted on a pair of 29.5-cm-long rods with mass 67.5 g each, as shown in Fig. 11.16. Find the angular momentum of the anemometer when it's spinning at 11.7 rev/s. You can treat the cups as point masses.

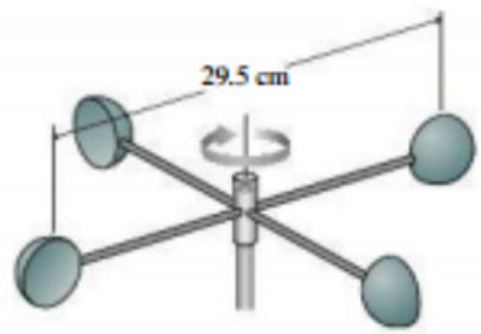

FIGURE 11.16 Problem 54